

RESEARCH ARTICLE

Unified gas kinetic scheme combined with Cartesian grid method for intermediate Mach numbers

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Funding information

Department of Science and Technology
(DST), New Delhi, India, Grant/Award
Number: IF140144

Summary

We develop a method to seamlessly simulate flows over a wide range of Knudsen numbers past arbitrarily shaped immersed boundaries. To achieve seamless computation, ie, not use any zone division to distinguish between continuum and non-continuum regions, we use the unified gas kinetic scheme (UGKS), which is based on the Bhatnagar-Groos-Krook (BGK) approximation of the Boltzmann equation. We combine UGKS with an appropriately designed Cartesian grid method (CGM) to allow us to compute flows past arbitrary boundaries. The CGM we use here satisfies boundary conditions at the wall by using a constrained least square interpolation procedure. However, it differs from the usual, continuum CGMs in 2 ways. Firstly, to allow us capture non-continuum effects at the boundaries, the CGM used herein interpolates the microscopic velocity distribution function in addition to the macroscopic variables. Secondly, even for the macroscopic variables, we use a gas kinetic method-based density interpolation procedure at the boundaries that allows the CGM to interface well with the UGKS method. We demonstrate the robustness and efficacy of the method by testing it on stationary immersed boundaries at various Knudsen numbers ranging from continuum to transition regimes.

KEYWORDS

BGK model, Cartesian grid method, least square interpolation, unified gas kinetic method

1 | INTRODUCTION

Flows involving both continuum and rarefied regimes occur in many practical situations such as flow past hypersonic aircrafts,¹ re-entry vehicles,² micronozzles for micropropulsion systems,³ thruster plume impingement,⁴ etc. Solving such flows requires the development of accurate numerical methods that can work seamlessly across the various flow regimes, viz, continuum, slip, transition, and rarefied. Traditional solvers such as the Direct Simulation Monte Carlo (DSMC)^{5,6} and modern CFD solvers based on the Navier-Stokes (NS) equations cannot be used for solving these flows as they face problems in one or the other regime. The DSMC is one of the most widely used techniques for simulating rarefied flows but encounters difficulties in simulating continuum and near-continuum flows. This is because, for low Knudsen (Kn) number flows (such as in the continuum regime), DSMC requires that the cell size and time step be less than the mean free path and collision time of the particles, respectively. These constraints impose enormous computational cost in the high-density regime for DSMC. On the other hand, conventional NS-based solvers cannot be used to capture nonequilibrium effects in rarefied flows as the continuum hypothesis breaks down in such cases. Higher-order hydrodynamic equations such as Burnett and Super Burnett equations are limited to solve flows up to transition regime only.⁷ In recent years, hybrid techniques that combines both the kinetic and the NS approaches have been

often used to model and simulate flows involving continuum and rarefied regimes.^{8–14} However, such techniques are based on the concept of zone division—the process of dividing the fluid flow domain into different regions depending on the flow regime. A buffer zone is used in such cases where it is assumed that both the kinetic and NS methods can be used. However, neither of these 2 techniques can be used with confidence in the buffer zone because of the limitations described above. Therefore, an alternative technique needs to be developed which can seamlessly simulate flows ranging from continuum to rarefied regime.

The Unified Gas Kinetic Scheme (UGKS)^{15–19} is a potential alternative to seamlessly simulate flows corresponding to different Knudsen number regimes. The UGKS is an extension of the conventional gas kinetic method (GKM)^{20,21} and can handle both equilibrium and non-equilibrium flows quite well. This ability of UGKS to compute flows in different regimes is because UGKS concurrently updates both the microscopic distribution function and the macroscopic variables at every time step. This allows UGKS to compute flows in rarefied regimes (where the distribution function is highly non-equilibrium), continuum regimes (where fluxes corresponding to the equilibrium part of the distribution function dominate), and the transition regime, (where fluxes from both the equilibrium and non-equilibrium distribution function play an important role).^{15,19} Moreover, in comparison to the operator splitting techniques such as the discrete velocity method^{2,22–24} and statistical techniques such as DSMC,^{5,6} the time step in UGKS is not restricted by the particle collision time. This is because UGKS couples both the particle transport and collision in the flux evaluation stage by using an analytical solution of the Boltzmann equation at each time step. Since these 2 physical phenomena are never decoupled in UGKS, the time step does not need to be smaller than the particle collision time. This brings a clear advantage for using UGKS in the continuum flow regime because, in such cases, the collision time between particles is very small on account of the large number of particle collisions. Finally, in case of the rarefied regime, the collision time between particles is large because of the larger intermolecular distances. Therefore, the time step based on the CFL condition can be easily chosen in both the continuum and rarefied flow regimes, and this allows UGKS to be used in seamlessly simulating different flow regimes.

Recently, UGKS has been applied to solve a wide variety of problems¹⁹ ranging from 1D to 3D flows, such as the study of shock structures, flows in microchannels, thermal transpiration flow, flow past bluff bodies such as square cylinders and circular cylinders, cavity flow in both 2D and 3D, slit flow etc. All the problems mentioned above are canonical flow problems and are easily solved using either a Cartesian mesh or a body conformal mesh. For complicated geometries, the use of an unstructured mesh is one of the most widely used approaches. However, unstructured meshes raise several difficulties within the UGKS framework. Firstly, it inherits the usual problems of generating the mesh. Secondly, extending the formulation of UGKS that involves phase space discretisation is not straightforward for unstructured meshes. Finally and most importantly, domain decomposition and parallelization would require complicated strategies for UGKS using unstructured meshes. These are probably the reasons that UGKS has not yet been developed for unstructured meshes in the literature to the best of our knowledge.

An alternate strategy, which has also gained increasing traction of late in the solution of continuum flows, is the use of immersed boundary method/Cartesian grid method (CGM).^{25–29} This involves the use of Cartesian meshes with appropriately defined corrections near the boundaries for imposing the desired boundary corrections. This allows for naturalness in parallelization at the cost of some work in devising correct schemes for representing the boundaries. Recently, Seo et al.³⁰ developed a CGM for continuum flows, which was adapted by Chen et al.,³¹ for computing compressible continuum flows by interfacing with conventional GKM.

In the present work, we further develop the above methods for compressible, noncontinuum flows. We demonstrate that UGKS can be combined with a suitably adapted CGM to compute incompressible and compressible flows past arbitrary bodies for a range of Knudsen numbers. We also show how the resultant UGKS-CGM can be parallelised using Message Passing Interface (MPI). Our purpose and primary achievement within this paper is, therefore, to show how one can handle arbitrary geometries and flows with a range of Knudsen numbers by combining UGKS and CGM. To this extent, our purpose is not to discover innovative algorithms within UGKS or continuum CGM but to show how CGM can be suitably modified for nonequilibrium flows so that one can compute flows with different geometries and Knudsen numbers.

The organisation of the rest of this paper is as follows. Section 2, presents a brief outline of the UGKS method. In Section 3, we develop a CGM for interfacing with UGKS. Section 4 presents several numerical test cases to validate the current method. Finally, in Section 5, we list the conclusions of this study and directions for future development.

2 | UNIFIED GAS KINETIC SCHEME

As mentioned Section 1, to achieve seamless computation for both continuum and non-continuum flows, we use the UGKS of Xu et al.^{15–19} In this section, we provide the essentials of the formulation for a 1-dimensional flow. The formulation given below

merely provides the framework required to understand the rest of the paper, and thus, for further details, the interested reader is referred to Xu and Huang¹⁸ and Wang and Xu.³²

The efficacy of the UGKS scheme is based on its concurrently updated solutions of the Boltzmann equation (for the microscopic and non-equilibrium distribution function) and the flow equations (for the macroscopic variables). For the former, we use the BGK-Shakhov model^{33,34} of the Boltzmann equation because of its ease in correcting for Prandtl number effects. In one dimension, with no external forces, the Boltzmann equation is given by

$$f_t + u f_x = \frac{f^+ - f}{\tau}, \quad (1)$$

where f is the gas distribution function, f^+ represents modified equilibrium distribution function, u is the microscopic velocity of the particles in x direction and τ is collision time, ie, the time interval between 2 successive particle collisions. The macroscopic conservation equations may be written as

$$W_t + F_x = 0, \quad (2)$$

where W represents the macroscopic, conservative variables. The above 2 equations are coupled by the macroscopic variables (mass, momentum, and energy densities) and are obtained from the microscopic distribution function f by taking moments with the collision invariants ψ_α , ie, by evaluating the phase space integral

$$W = \begin{pmatrix} \rho \\ \rho U \\ \rho e \end{pmatrix} = \int \begin{bmatrix} 1 \\ u \\ \frac{u^2}{2} \end{bmatrix} f du d\xi. \quad (3)$$

The corresponding fluxes are given by

$$F(W) = \begin{pmatrix} F_\rho \\ F_{\rho U} \\ F_{\rho e} \end{pmatrix} = \int u \begin{bmatrix} 1 \\ u \\ \frac{u^2}{2} \end{bmatrix} f du d\xi. \quad (4)$$

The collision time between particles τ is related to dynamic viscosity of fluid μ , and pressure p by $\tau = \frac{\mu}{p}$. Finally, the modified equilibrium distribution function f^+ has a one-to-one correspondence with the macroscopic variables given by

$$f^+ = g \left[1 + (1 - Pr) \mathbf{u} \cdot \mathbf{q} \left(\frac{u^2}{RT} - 5 \right) / (5pRT) \right] = g + g^+, \quad (5)$$

where g is the equilibrium distribution function, $\mathbf{q}$ is the heat flux, Pr is the Prandtl number, R is the gas constant, and T is the temperature. The equilibrium distribution function g in Equation 5 is the Maxwellian, given by

$$g = \rho \left(\frac{1}{2\pi RT} \right)^{\frac{1}{\gamma-1}} e^{-\frac{1}{2RT}[(u-U)^2]}. \quad (6)$$

The finite volume implementation of UGKS involves the discretisation of physical and temporal space as usual. In addition, the phase space (microscopic particle velocity space) is also discretised. The centre of the k th microscopic velocity interval represents the average velocity of particle in that interval

$$u_k \in \left[\left(k - \frac{1}{2} \right) \Delta u_k, \left(k + \frac{1}{2} \right) \Delta u_k \right]; \quad k = -M, -(M-1), \dots, -1, 0, 1, \dots, (M-1), M.$$

Accordingly, the average of gas distribution function in cell j , at time step t^n , and around particle velocity u_k , is given by

$$f(x_j, t^n, u_k) = f_{j,k}^n = \frac{1}{\Delta x \Delta u} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \int_{u_{k-\frac{1}{2}} \Delta u}^{u_{k+\frac{1}{2}} \Delta u} f(x, t^n, u) dx du \quad (7)$$

With the discretisation of space x_j , time t^n , and particle velocity u_k , the finite volume scheme based on integral solution of Equation 1 is

$$f_{j,k}^{n+1} = f_{j,k}^n + \frac{1}{\Delta x} \int_{t^n}^{t^{n+1}} \left(u_k \hat{f}_{j-\frac{1}{2},k}(t) - u_k \hat{f}_{j+\frac{1}{2},k}(t) \right) dt + \frac{1}{\Delta x} \int_{t^n}^{t^{n+1}} \int_{x_{j-\frac{1}{2}}}^{x_{j+\frac{1}{2}}} \frac{f^+ - f}{\tau} dx dt \quad (8)$$

where $f_{j,k}^n$ is the particle distribution function in j^{th} cell at particle velocity u_k and time level t^n . Similarly, Equation 2 governing the macroscopic variables can be discretised by taking moments of the collision invariants with Equation 8. Then, the evolution equation for the conservative flow variables becomes

$$W_j^{n+1} = W_j^n + \frac{1}{\Delta x} \int_{t^n}^{t^{n+1}} u (\tilde{g}_{j-\frac{1}{2}} - \tilde{g}_{j+\frac{1}{2}}) \psi dt du d\xi + \frac{1}{\Delta x} \sum_k \int_{t^n}^{t^{n+1}} u_k (\tilde{f}_{j-\frac{1}{2}} - \tilde{f}_{j+\frac{1}{2}}) \psi dt. \quad (9)$$

In Equation 9, $\tilde{g}_{j\pm\frac{1}{2}}$ and $\tilde{f}_{j\pm\frac{1}{2}}$ represent the equilibrium and non-equilibrium distribution functions at the cell face, respectively. Applying the trapezoidal rule for time integration of the collision term in Equation 8, the UGKS for update of gas distribution function becomes

$$f_{j,k}^{n+1} = \left(1 + \frac{\Delta t}{2\tau_j^{n+1}}\right)^{-1} \left[f_{j,k}^n + \frac{1}{\Delta x} \left(\int_{t^n}^{t^{n+1}} u_k \left(\tilde{g}_{j-\frac{1}{2},k} - \tilde{g}_{j+\frac{1}{2},k} \right) dt + \int_{t^n}^{t^{n+1}} u_k \left(\tilde{f}_{j-\frac{1}{2},k} - \tilde{f}_{j+\frac{1}{2},k} \right) dt \right) + \frac{\Delta t}{2} \left(\frac{g_{j,k}^{n+1}}{\tau_j^{n+1}} + \frac{g_{j,k}^n - f_{j,k}^n}{\tau_j^n} \right) \right] \quad (10)$$

In Equation 10, τ_j^n and τ_j^{n+1} represent the collision time at time level n and $n+1$, respectively. They are calculated using the temperatures and pressures in the cells, ie, $\tau_j^{n+1} = \frac{\mu(T_j^{n+1})}{P_j^{n+1}}$, which are known because of the update of the macroscopic variables in Equation 9.

The updates of the macroscopic variables and the microscopic distribution function using Equations 9 and 10, respectively, allow UGKS to compute both equilibrium and non-equilibrium flows accurately. The above formulation can be easily extended to 2D and 3D versions by extending Equations 1, 9, and 10 in the additional directions.^{15,19}

3 | THE UGKS-CGM

The UGKS method described in the Section 2 is capable of seamlessly simulating flows across a wide range of Knudsen numbers. However, it needs further modification to be capable of handling practical flows involving arbitrary geometries. Since the cost of the underlying UGKS scheme is quite high (owing to its 6-dimensional phase space), parallelization is essential for computing practical flows. Unstructured mesh methods do not adapt themselves as naturally to parallelization as CGMs. For this reason, in the rest of this work, we choose to combine UGKS with a suitably derived CGM to compute flows past arbitrary boundaries over a wide range of Knudsen numbers.

The use of Cartesian meshes for non-conforming boundaries was pioneered by Peskin²⁵ to simulate cardiac mechanics and the associated blood flow. Since then, a large number of variants have been developed within the larger framework of CGMs. The primary challenge in every variant is the enforcement of the boundary condition at the non-conforming boundary (often labelled the immersed boundary).

We refer the reader to the excellent review by Mittal and Iaccarino³⁵ for a survey of the various methods within the larger class of CGMs. We focus below on the variants that are relevant to our desired application—computing flows for various speeds and Knudsen numbers past rigid bodies. Accordingly, we make the following choices regarding our CGM—

- Since we are interested primarily in flow past rigid bodies, we use the discrete forcing approach rather than the continuous forcing approach,^{36–38} as the latter does not behave well in the rigid limit.
- Within the discrete forcing approach, we directly impose the boundary conditions and treat the boundary as a “sharp” (as opposed to diffuse) interface since we are interested in accurate solutions near the boundaries.
- We choose to apply the boundary conditions strongly rather than weakly as UGKS enforces the additional requirement of imposing boundary conditions for the distribution function at the interface, and there is no simple criterion by which one can apply this weakly. Further, weak boundary conditions do not give us any added computational advantage since the predominant cost of the method is driven by UGKS rather than the CGM.
- We choose to use a ghost cell method rather than a cut cell method²⁹ as the latter is very complicated to implement as well as harder to parallelise, especially when extended to 3D.

3.1 | The ghost cell approach

In the ghost cell approach, boundary conditions on the immersed boundary are enforced with the help of ghost cells, which are defined as the cells in the solid domain that have at least 1 neighbour in the fluid region. For each ghost cell, an interpolation scheme is used to calculate the flow variables in such a way that the boundary conditions are exactly satisfied at the boundary. This procedure leads to a sharp representation of the boundary. An early application of this technique was done by Forrer and Jeltsch³⁹ to simulate inviscid compressible flow. In their approach, the flow variables at the ghost cells are calculated by using their images in the fluid side across the boundary. Since the fluid cells adjacent to the boundary rely only on the ghost cells for

their fluxes, there is no need to keep track of the intersection of grid lines and the boundary that plays a key role in the cut cell approach.

The above method was extended by Seo and Mittal in a study³⁰ to compute low Mach number flow-induced sound around complex geometries. In their method, they combined the ghost cell approach with a weighted least square error method on the basis of a higher order approximating polynomial to specify flow quantities in the ghost cells. Recently, Chen et al²⁸ extended this further to simulate both incompressible and compressible viscous flows by combining the CGM with a gas kinetic solver. They used a modified gas kinetic scheme (GKS),³¹ which is based on the conventional GKS developed by Xu et al.^{20,21,40} Unlike UGKS, conventional GKS does not compute the distribution function at each time step. Instead, the deviation of non-equilibrium distribution function from the equilibrium distribution function is modelled using the Chapman Enskog expansion^{21,31} at each time step. Therefore, such solvers can compute flows only in the near equilibrium regime, and hence, the combination of CGM with simplified GKM²⁸ is limited to solve NS flows, ie, flows with low Knudsen numbers only.

In the present work, we focus on removing the above mentioned limitation of Chen's CGM.³¹ We aim to develop a solver that is not only able to solve the flow past complicated geometries (this is the primary purpose for which CGM is actually used) but also cover a wide range of Knudsen numbers (the most important feature of the UGKS scheme). Therefore, we focus on developing a UGKS-CGM solver such that the CGM interfaces accurately with UGKS. As with typical ghost cell CGM approaches, our CGM also involves 2 distinct steps:

1. Stencil creation: In this step, we create an interpolation stencil and classify the points into 3 different categories—fluid points (FPs), solid points (SPs), and the interpolation points (IPs).
2. Interpolation: In the second step, we interpolate the physical variables at the IPs using a suitable interpolation procedure.

In both these steps, our interfacing with UGKS requires that we have certain points of divergence from our precursors such as Chen et al²⁸ and Seo and Mittal.³⁰ We highlight these differences at the relevant places while explaining the implementation details of each of these steps in the subsections below.

3.2 | Creation of interpolation stencil

Figure 1 illustrates the interpolation stencil used for interpolating all the physical variables (macroscopic and microscopic) at the IPs. The steps required for creating this interpolation stencil are mentioned below.

1. We define the solid boundary by providing the coordinates of points lying on the boundary (boundary points). The boundary points are represented by stars in Figure 1.
2. We create a computational domain and mesh the domain using a Cartesian mesh. The mesh maybe uniform or nonuniform.

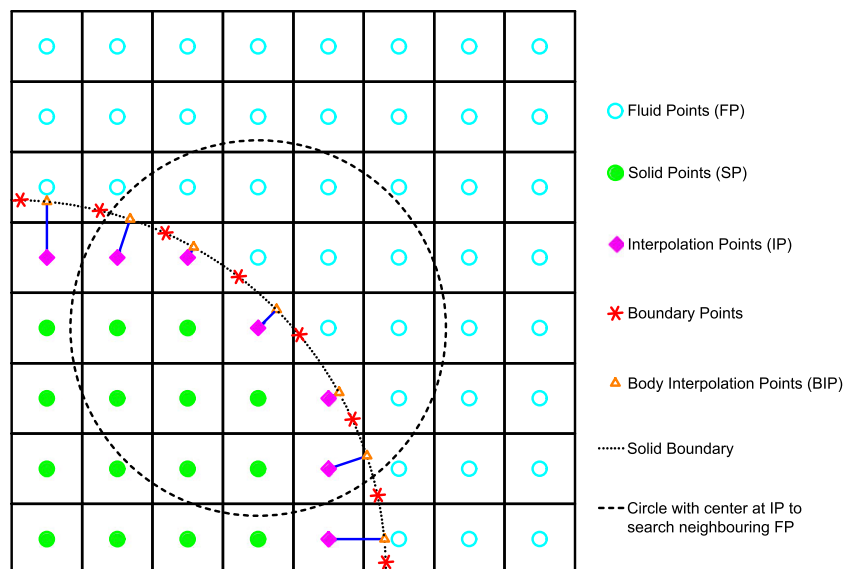


FIGURE 1 The stencil for least square interpolation [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/fld.4393)]

3. We classify the grid points as either SPs or FPs. The grid points that lie inside the solid boundary are marked as SPs (represented by filled circles in Figure 1). Those grid points that lie outside the solid boundary are marked as FPs (represented by unfilled circles in Figure 1). For deciding whether a grid point lies inside or outside of solid boundary, we use a ray tracing algorithm.^{41,42}
4. We identify the IPs, ie, the points where all physical variables will be interpolated in such a manner that the boundary conditions are exactly satisfied at the boundary. Any SP that has a fluid neighbour next to it gets converted into an IP (represented by filled diamonds in Figure 1).
5. We create a normal from each IP to the boundary and find the point of intersection which is labelled the body interpolation point (BIP). Depending on the shape of the immersed body, we may have multiple BIPs for a single IP. From this set of BIP and for a given IP under consideration, we select the one that is nearest to the IP. Thus for each IP, we have only 1 BIP associated with it, and this BIP acts as the point where the constraint of satisfying the boundary conditions will be imposed in the interpolation procedure. In Figure 1, the body intercept points are represented as triangles and normal from an IP to the boundary is shown by a line connecting an IP to a BIP.
6. We create an interpolation circle of radius R at each IP (the criteria for choosing R are given in Section 3.3). The FPs that fall inside the circle are the points that are used in the interpolation procedure mentioned in Section 3.3.

In summary, for any given IP, the interpolation stencil consists of all the FPs within the corresponding interpolation circle, and the single BIP. The interpolation stencil chosen here is similar to the one developed by Chen et al³¹ but differs in one important respect. This is in the definition of the BIPs that are finally used for imposing boundary conditions. In Chen's method, the boundary points (which are used to define the shape of the immersed solid object) double as the points where the constraints of satisfying the boundary conditions are implemented. We, on the other hand, use normals that are drawn from the IP to define the corresponding BIPs and then use this BIP as the point where the constraints of satisfying the boundary conditions are implemented. We make this choice primarily because of our implementation of the UGKS density boundary condition that requires these points (see Section 3.3.2). We have also found this method to add some robustness to the interpolations.

3.3 | The interpolation step

Once the interpolation stencil is determined, we use it to determine the flow variables at the ghost cell centre (which we label the IP). The primary challenges here arise because of our need to interface the method with UGKS. Therefore, just like the basic UGKS scheme (without CGM), we first require the value of distribution function at the wall (more specifically, we require the values of the distribution function at the BIP).

There are several choices for this interpolation at the BIP. Firstly, we could interpolate the macroscopic variables and then use the corresponding equilibrium function. Alternately, one could also use a low-order Chapman-Enskog expansion from the macroscopic variables. Finally, we could also interpolate the full distribution at the wall by using the same interpolation procedure that we use to interpolate the macroscopic variables. Our numerical test showed that only the full interpolation works well with UGKS.

This interpolation of the distribution function is unique to UGKS-CGM because UGKS heavily relies on the values of distribution function at every time step. This also has downstream effects on the interpolation of macroscopic variables as well. The no penetration boundary condition at the immersed boundary can be consistently applied only if the interpolated distribution function is used to derive the density. Therefore, once we have the distribution function at the wall, we calculate the density at BIP by applying the no penetration boundary condition along the normal that was drawn from the IP to its corresponding BIP. The density thus obtained combined with the velocity and temperature boundary conditions give all the flow variables at the boundary. The boundary variables thus obtained are then used as constraints to evaluate the flow quantities at the IPs. The overall interpolation process, therefore, consists of the following 3 steps.

For every IP (ie, ghost point),

1. we first interpolate the distribution function at the BIP using the stencil values at the corresponding FPs using an unconstrained least squares procedure;
2. we calculate the density at the BIP using a consistent no penetration boundary condition, which is detailed below;
3. finally, we interpolate the conservative variables and the distribution function to the IP (ie, ghost point) while enforcing the desired boundary condition using a constrained least squares procedure.

The whole procedure differs from Chen et al³¹ significantly because of the interpolation of the distribution function and the corresponding interpolation of the density. We now explain each of these steps in the subsequent sections given below.

3.3.1 | Interpolation of the distribution function at the BIP

We first assume that the fluid variables near the boundary can be expressed by a polynomial

$$P(x, y) = C_1 + C_2x + C_3y, \quad (11)$$

where x and y are the coordinates of the point where we have to find the value of the fluid variable. In Equation 11, C_1 , C_2 , and C_3 are the unknown coefficients that need to be determined. We store these unknown coefficients in a vector $\mathbf{X}$, which can be represented as

$$\mathbf{X} = (C_1, C_2, C_3)^T. \quad (12)$$

We draw a circle of radius R at any given IP such that there are N neighbouring FPs that fall inside the circle. The radius of the circle should be such that $N \geq \dim(\mathbf{X})$. Let $\{v_i\}$ be the values of physical variables which are known at the set of locations $\{(x_i, y_i) \mid i = 1, \dots, N\}$. Here v_i denotes every single value of W_i and every single value of the distribution function $f_{m,n}$ where $1 \leq m \leq p$ and $1 \leq n \leq q$. p and q represent the number of velocity points for the microscopic particle velocity spaces u and v respectively. Let $\mathbf{A}$ be a matrix of size $(N, \dim(\mathbf{X}))$, which stores the coordinates of all the FP neighbours of an IP. The i th row of the matrix $\mathbf{A}$ is given by

$$\mathbf{A}_i = (1, x_i, y_i). \quad (13)$$

In Equation 13, x_i, y_i are the coordinates of the i th neighbouring FP. Let $\mathbf{v}$ be a vector given by $\mathbf{v} = (v_1, v_2, \dots, v_N)$. The objective of the least square interpolation is to minimise a mean square error function, which reads

$$\text{minimise } \sum_{i=1}^N (\mathbf{A}_i \mathbf{X} - v_i)^2. \quad (14)$$

The solution of the above unconstrained minimisation problem can be given as

$$\mathbf{X} = \mathbf{C}^{-1} \mathbf{A}^T \mathbf{v}, \quad (15)$$

where $\mathbf{C} = \mathbf{A}^T \mathbf{A}$. Using Equations 15 and 12, we can interpolate the distribution function at the BIP corresponding to the IP under consideration by

$$f_{\text{BIP}}^{\text{interp}} = C_1 + C_2 x_{\text{BIP}} + C_3 y_{\text{BIP}}, \quad (16)$$

where $f_{\text{BIP}}^{\text{interp}}$ is the interpolated distribution function at the BIP. In Equation 16, x_{BIP} and y_{BIP} are the coordinates of the BIP where the distribution function is to be interpolated.

3.3.2 | Density calculation at BIP using no penetration boundary condition

Once we have the non-equilibrium distribution function at the BIP, we need to calculate the density there. To be consistent with the condition that no particle penetrates the wall, we need to ensure that the net mass flux normal to the wall vanishes. For consistency with the interpolated distribution function, this requires that the following condition be applied along the normal direction from IP to BIP

$$\int_{t^n}^{t^{n+1}} \int_{u>0} u g_w du \xi dt + \int_{t^n}^{t^{n+1}} \int_{u<0} u f^{\text{interp}} du d\xi dt = 0, \quad (17)$$

which gives

$$\rho_w = - \frac{\sum \alpha_k u_k f_{\text{BIP},k}^{\text{interp}}}{\left(\frac{\lambda_w}{\pi}\right) \sum \alpha_k u_k e^{-\lambda_w((u-U_w)^2+(v-V_w)^2)}}, \quad (18)$$

where g_w , ρ_w , and λ_w are the variables at wall. The variables u_k and v_k represent microscopic velocities normal and parallel to the wall, respectively. Similarly, U_w and V_w represent the macroscopic velocities normal and parallel to the wall, respectively. The symbols α_k represents the weights used in the numerical integration.

Using the density obtained in Equation 18 and using the values of U_w , V_w , λ_w (known from the boundary conditions), we can obtain equilibrium distribution function g_w at the wall by

$$g_w = \rho_w \left(\frac{\lambda_w}{\pi}\right)^{\frac{3}{2}} e^{-\lambda_w((u-U_w)^2+(v-V_w)^2+\xi^2)}. \quad (19)$$

Finally, we calculate the nonequilibrium distribution function at the BIP by

$$f_{BIP} = g_w H[u_k] + f_{BIP}^{interp} (1 - H[u_k]), \quad (20)$$

where $H[u_k]$ is the Heaviside step function. From Equations 20 and 18, we get both the microscopic distribution function (f_{BIP}) at the wall and all 4 primary variables (ρ_w, U_w, V_w , and λ_w) at the boundary.

3.3.3 | Interpolation of conservative variables and distribution function at the IP

From the 4 primary variables ρ_w, U_w, V_w , and λ_w obtained in Section 3.3.2 and the equilibrium distribution function calculated in Equation 19, the macroscopic conservative variables W_{BIP} at the BIP are calculated by

$$W_{BIP} = \begin{pmatrix} \rho \\ \rho U \\ \rho V \\ \rho e \end{pmatrix} = \int \begin{pmatrix} 1 \\ u \\ v \\ \frac{1}{2}(u^2 + v^2 + \xi^2) \end{pmatrix} g_w d u d \xi. \quad (21)$$

We now use f_{BIP} obtained in Equation 20 and W_{BIP} obtained in Equation 21 as constraints to interpolate the microscopic distribution function and the macroscopic conservative variables at the interpolation (ghost) point. The coefficient vector defined in Equation 12 should be such that it must satisfy the constraint:

$$\mathbf{A}_b \mathbf{X} = \mathbf{v}_b, \quad (22)$$

where v_b denotes every single value of W_{BIP} and f_{BIP} . The symbol A_b in Equation 22 is a row vector containing the coordinates of the BIP given by

$$\mathbf{A}_b = (1, x_{BIP}, y_{BIP}). \quad (23)$$

Combining the constraint given in Equation 22 with Equation 14, the constrained solution is

$$\mathbf{X} = \mathbf{C}^{-1}(\mathbf{I} - \mathbf{A}_b^T \mathbf{C}_b^{-1} \mathbf{A}_b \mathbf{C}^{-1}) \mathbf{A}^T \mathbf{v} + \mathbf{C}^{-1} \mathbf{A}_b^T \mathbf{C}_b^{-1} \mathbf{v}_b, \quad (24)$$

where

$$\mathbf{C}_b = \mathbf{A}_b \mathbf{C}^{-1} \mathbf{A}_b^T. \quad (25)$$

Thus, we fully determine the polynomial $P(x, y)$ at the fluid side. Using the values of $\mathbf{X}$ obtained in Equations 24 and 11, we calculate the flow variables at the IP by

$$P(x_{IP}, y_{IP}) = C_1 + C_2 x_{IP} + C_3 y_{IP}, \quad (26)$$

where x_{IP}, y_{IP} are the coordinates of the IP. Using Equations 24 and 26, we can now obtain both the microscopic distribution f_{IP} and the macroscopic conservative variables W_{IP} . This completes the interpolation procedure at the IP. The whole UGKS-CGM procedure may now be summarised as follows. For every time step,

1. update the conservative variables using Equation 9 at all FPs,
2. update the distribution function using Equation 10 at all FPs,
3. determine the IPs and create the respective stencils according to the procedure outlined in Section 3.2,
4. perform the unconstrained least square interpolation for the distribution function at the BIPs as given in Section 3.3.1,
5. perform the density calculation at the BIPs as given in Section 3.3.2, and
6. update the variables in the ghost cells (IPs) using the constrained least squares procedure as given in Section 3.3.3.

4 | RESULTS AND DISCUSSIONS

The above UGKS-CGM method was tested extensively for a range of cases to check the accuracy of our solver in different flow regimes (continuum, slip, and transition regime). We considered the following cases:

1. Translated lid-driven cavity—This is intended to represent the simplest case where the mesh lines do not coincide with lines representing the boundary.

2. Rotated lid-driven cavity—This case presents a more complicated version of the above where the mesh lines are at an angle to the boundary. This tests the basic efficacy of the method across various Knudsen number regimes as well.
3. Polar lid-driven cavity—This tests for completely nonaligned boundaries in internal flows. Further, the boundary has a rotational velocity component.
4. Flow past a circular cylinder—This tests for a non-Cartesian boundary in external incompressible and moderately compressible flows. This flow is also tested for several Knudsen numbers.
5. Flow past a NACA 0012 airfoil—This tests for a non-Cartesian boundary in an external compressible flow.

Finally, we discuss the determination of the interpolation radius (which is used to find the interpolation stencil) in Section 4.5. In all the test cases described below, we use the Argon gas with Prandtl number $Pr = 2/3$ and $\omega = 0.81$ corresponding to the Variable Hard Sphere (VHS) model.

4.1 | Translated and rotated lid-driven cavity

The lid driven cavity case is a standard benchmark case for incompressible flows.⁴³ Following Chen et al,²⁸ we use translated and rotated versions of this case to test the efficacy of our CGM for nonaligned boundaries. We also show that our UGKS-CGM

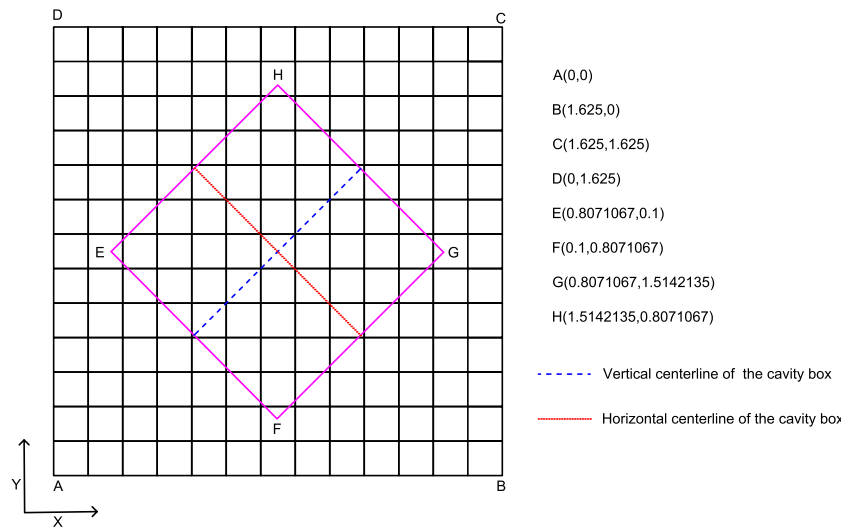


FIGURE 2 Rotated lid-driven cavity (diagram not to scale) [Colour figure can be viewed at wileyonlinelibrary.com]

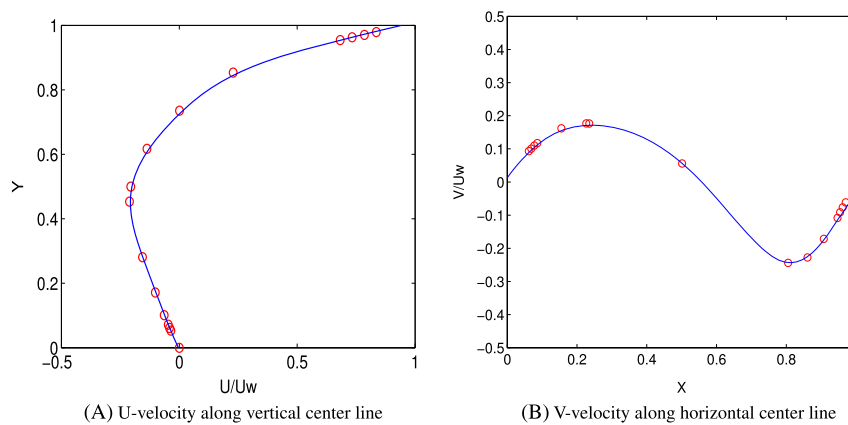


FIGURE 3 Continuum lid-driven cavity flow at $Re = 100$. $\circ$: Ghia's data from Ghia et al⁴³ and —: unified gas kinetic scheme-Cartesian grid method simulation results [Colour figure can be viewed at wileyonlinelibrary.com]

works effectively not only for continuum flows but also for higher Knudsen numbers. For brevity, we only present the results for the rotated cavity case as the translated cavity case results were similar in accuracy.

The rotated lid driven cavity is shown in Figure 2. Here, the cavity is rotated at 45° with respect to the mesh lines. The cavity wall GH moves at a constant wall velocity such that $U_w = V_w = U_w^t \cos(45^\circ)$, where U_w^t represents the magnitude of wall velocity along GH . All other walls are stationary, and all the walls are maintained isothermally at $T = 1$. We tested the above case at different Knudsen numbers corresponding to continuum, slip, and transition regimes. The steady state velocities at the vertical and horizontal centrelines of the cavity were compared with standard solvers. The details of our simulations are given below. In all the cases, the Knudsen number was calculated by using

$$Kn = \frac{Ma}{Re} \sqrt{\frac{\gamma\pi}{2}}, \quad (27)$$

where required Mach and Reynolds numbers were imposed to obtain the desired Knudsen number ranges. The cases we tested for are

1. Continuum regime: Continuum flow requires $Kn < 0.01$. We set $Re = 100$ and $M = 0.3$. The corresponding $Kn = 0.004854$, which is in the continuum regime. Physical space was discretised using a mesh of 265×265 points. For discretisation of the velocity (phase) space, 28 points were used in each direction and Gauss Hermite quadrature was used to evaluate the moments of distribution function. In Figure 3, we plot the steady-state wall velocity at horizontal and vertical centreline of the cavity box. The simulation results are compared with those of Ghia et al.⁴³

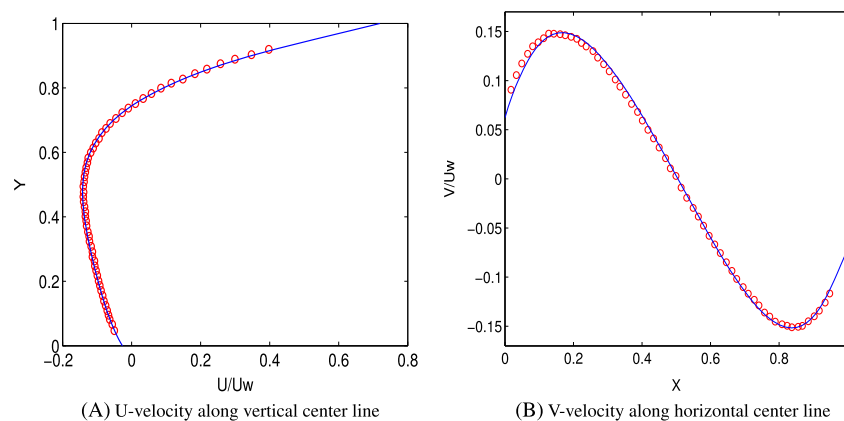


FIGURE 4 Slip lid-driven cavity flow at $Kn = 0.075$. $\circ$: Direct simulation Monte Carlo data from Huang et al.¹⁵ and —: unified gas kinetic scheme-Cartesian grid method simulation results [Colour figure can be viewed at wileyonlinelibrary.com]

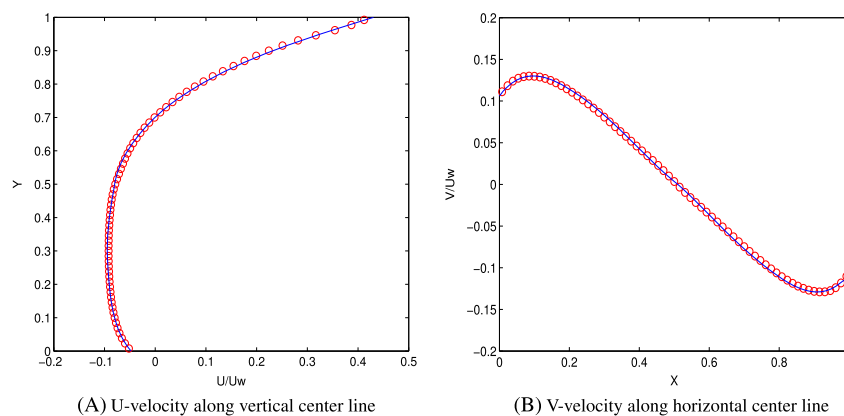


FIGURE 5 Transition lid-driven cavity flow at $Kn = 1.0$. $\circ$: Unified gas kinetic scheme simulation (UGKS) results (without Cartesian grid method) and —: UGKS-Cartesian grid method simulation results [Colour figure can be viewed at wileyonlinelibrary.com]

2. Slip flow regime: Slip flow requires $0.01 < Kn < 0.1$. This case had $Kn = 0.075$ and $U_w = 0.15$. The computational domain was discretised using 200×200 points in the physical space and 28×28 points in the velocity space. Moments of the distribution function are calculated using Gauss Hermite quadrature. Figure 4 shows the simulation results at steady state. The simulation results are compared with the DSMC results from Huang et al.¹⁵
3. Transition regime: Transition flow is in the range $0.1 < Kn < 10$. This case had $Kn = 1.0$ and $U_w = 0.15$. The computational domain was discretised using 151×151 points in the physical space and 101×101 points in the velocity space. Moments of the distribution function are calculated using Newton-Cotes quadrature. The particle velocity space varies in the range $[-5.0, 5.0]$ for both u and v velocities. Figure 5 shows the simulation results at steady state. The simulation results are compared with the DSMC results from Huang et al.¹⁵

An inspection of these results confirms that the UGKS-CGM solver works very well for a range of Knudsen numbers from the continuum to the transition regime.

4.2 | Polar lid-driven cavity

The cases in the previous section involve simple boundary geometries. To check the performance of our UGKS-CGM solver for simulating internal flows that involve more complex geometries, we consider the polar lid-driven cavity. The problem geometry consists of a wall bounded region of an annulus with an inner rotating wall (Figure 6). The inner circular wall rotates in the

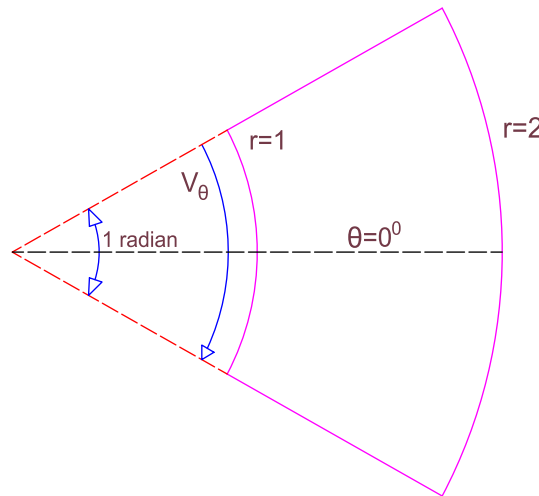


FIGURE 6 Schematic diagram of polar lid-driven cavity problem [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

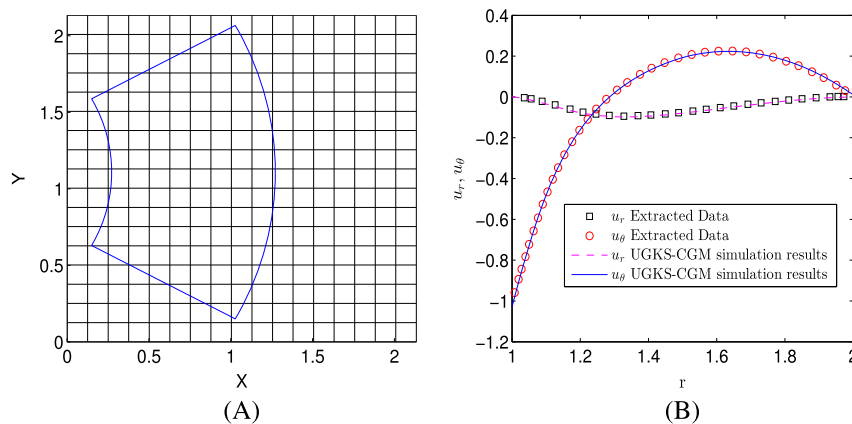


FIGURE 7 A, A portion of the meshed computational domain for polar lid-driven cavity problem. B, Radial and azimuthal velocity distribution for $Re = 55$ at $\theta = 0^\circ$. For comparison, data shown with symbols has been extracted from Tavakoli et al.⁴⁴ CGM, Cartesian grid method; UGKS, unified gas kinetic scheme [Colour figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

clockwise direction at a constant angular velocity (V_θ) with magnitude equal to 1.0 providing additional complexity for the boundary condition. All lengths are normalised by the radius of the inner wall (r_i), and velocities are normalised by the angular velocity of the inner wall (V_θ). The Reynolds number is defined on the basis of inner wall velocity and radius and is given by $Re = V_\theta r_i / \nu$. The Reynolds number for the current simulation is the same as mentioned in Tavakoli et al.⁴⁴ and is equal to 55. The Mach number for the simulation is chosen to be 0.3 to keep the simulation flow field in the incompressible limit.

The computational domain is a square box as shown in Figure 7A. The length of the computational domain is 2.2177 in each direction. The physical space is discretised using a mesh of 265×265 points in x and y directions. The velocity space consists of 28 points each for u and v velocities.

In Figure 7B, we plot the radial and azimuthal velocities at the horizontal centreline of the domain ($\theta = 0^\circ$). We compare our simulation results with those of Tavakoli et al.⁴⁴ We observe that there is an excellent agreement between the UGKS-CGM simulation results and Tavakoli's results proving that the UGKS-CGM solver can solve flows involving moderately complex geometries quite accurately.

4.3 | Flow past circular cylinder

In this test case, we consider an external flow over a circular cylinder. This test case is often used as a benchmark for non-continuum CGM-based solvers as well. We test this geometry for both continuum and slip flow cases. The details of the cases are as given below.

In both cases, our computational domain is a rectangular box with the inlet and outlet vertical boundaries at $x = -15D$ and $x = 15D$ as shown in Figure 8. The 2 horizontal far field boundaries are located at $y = -15D$ and $y = 15D$. The cylinder is placed inside the domain with its centre at the origin.

At the inlet of the computational domain, we use the following boundary conditions

$$U = \text{Constant}, \quad V = 0.0, \quad T = 1.0, \quad \rho = 1.0, \quad (28)$$

where U velocity is calculated using the value of Reynolds number. At the other 3 boundaries, we impose zero normal derivatives on the horizontal velocities, temperature, and density. The vertical velocities are set to 0 on the top and bottom boundaries, and its derivative is set to 0 at the right boundary (ie, the outlet).

For both cases, the computations were performed on a non-uniform grid consisting of 302×302 points in the physical space and 28×28 points in the velocity space. The cell size varies according to a geometric progression with minimum cell size at the domain centre and a growth ratio of 1.017 (Figure 8). We ran our code (parallelised using MPI) on E5-2680 v3 Intel Xeon Processors consisting of 12 cores with a clock rate of 2.5 GHz. The simulation is run on a total of 25 cores with a 2D tile arrangement of 5 cores in each direction. We performed our simulations for flows in both the continuum and slip regimes as detailed below.

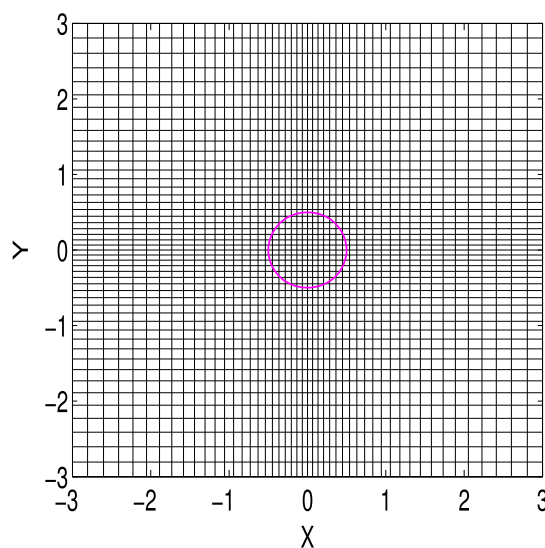


FIGURE 8 A portion of the meshed computational domain for flow past cylinder (1 in every 3 mesh lines is shown for clarity) [Colour figure can be viewed at wileyonlinelibrary.com]

4.3.1 | Continuum regime

For this, we select a test case from Dennis and Chang⁴⁵ with $Re = 20$. We note that Dennis and Chang simulated an incompressible flow using incompressible NS solver. However, since UGKS is inherently formulated for compressible flows, we choose a low value of Mach number (0.111) to ensure that our simulation flow field is close to the incompressible limit. The corresponding Knudsen number of our simulated flow field is found to be 0.009, which is low enough to be considered in the continuum regime.

In Figure 9A, we plot the contours of streamwise velocity at steady state. We also evaluate the results of our simulation of the coefficient of pressure (C_p) on the cylinder wall. In Figure 9B, we present the variation of C_p versus θ , where θ represent locations on the cylinder wall. The forward stagnation point has $\theta = 0^\circ$ and the rear of cylinder has $\theta = 180^\circ$.

For comparison, we have included results from Dennis and Chang.⁴⁵ Evidently, the agreement between the incompressible NS solver and our UGKS-CGM is excellent. This confirms that the UGKS-CGM method can capture flow past 2D bluff bodies in the continuum regime quite well.

4.3.2 | Slip flow regime

To examine the applicability of the UGKS-CGM solver in the slip regime, we consider test cases from Hu et al.⁴⁶ at $Re = 10$. Following Hu et al.,⁴⁶ we choose 2 different Mach numbers of 0.2 and 0.6, respectively, representing nearly incompressible and moderately compressible flows. The corresponding Knudsen numbers for the simulation are 0.032 and 0.097, both of which are in the slip regime.

In Figure 10A and 10B, we plot the variation of C_p versus θ at Mach numbers of 0.2 and 0.6, respectively. We compare our simulation results with the DSMC results of Hu et al.⁴⁶ Evidently, the agreement is quite good for both the Mach numbers.

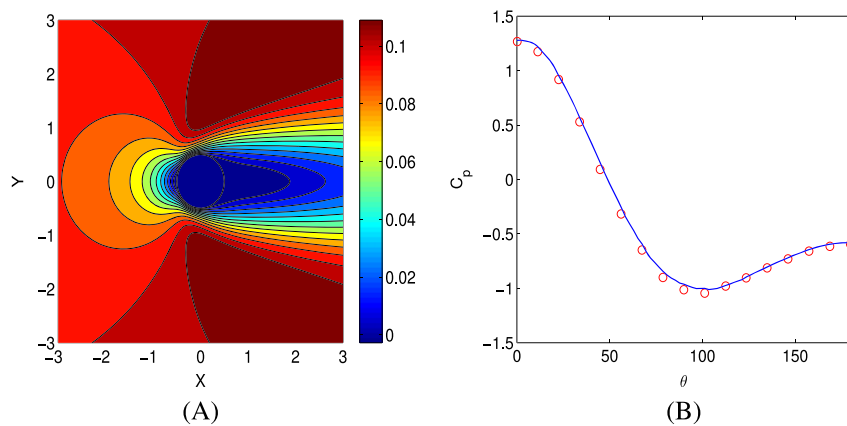


FIGURE 9 Simulation results for flow past circular cylinder. A, Streamwise velocity contours from the current unified gas kinetic scheme (UGKS)-Cartesian grid method (CGM) simulation results. B, C_p versus θ . $\circ$: Dennis's data⁴⁵ and —: UGKS-CGM simulation results [Colour figure can be viewed at wileyonlinelibrary.com]

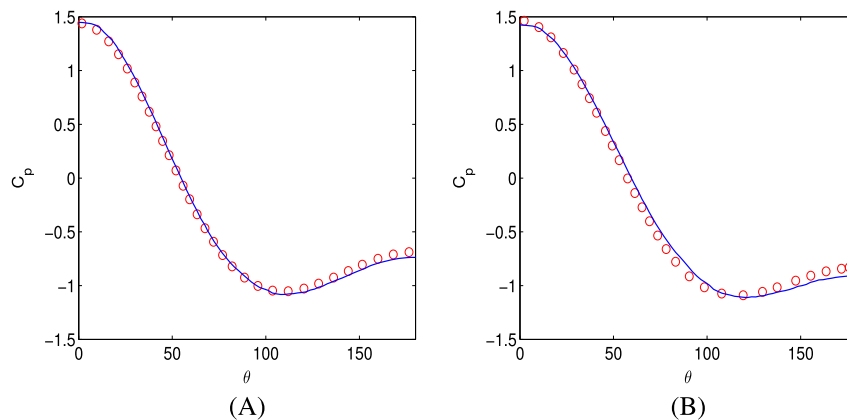


FIGURE 10 C_p versus θ for the flow past a circular cylinder in the slip flow regime at A, $Re = 10$ and $Ma = 0.2$ and B, $Re = 10$ and $Ma = 0.6$. $\circ$: Direct simulation Monte Carlo results from Hu et al.⁴⁶ and —: Unified gas kinetic scheme-Cartesian grid method simulation results [Colour figure can be viewed at wileyonlinelibrary.com]

This demonstrates that the UGKS-CGM solver is capable of simulating incompressible and compressible slip flows. These cases also demonstrate that the method can accommodate non-uniform meshes as well.

4.4 | Flow past NACA 0012 airfoil

To further evaluate the UGKS-CGM solver for arbitrary geometries, we consider a test case from Fan et al.⁴⁷ The test case consists of simulating the flow past a NACA 0012 airfoil in the slip regime. The setup of the problem and its parameters are kept the same as mentioned in Fan et al.⁴⁷ Chord length of the airfoil C is equal to $0.04 \mu\text{m}$. The values of the Reynolds and Mach number used in the simulation are 73.0 and 0.8, respectively.

The airfoil is placed at the domain centre at zero angle of attack. The computational domain and the boundary conditions for this case are identical to that given in Section 4.3. However, the mesh is somewhat different. A fine horizontal and a vertical zone is defined in the near airfoil region (Figure 11). The uniform mesh covers a region of $0.55C$ in each direction from the domain centre. A total of 400 points are used for creating the uniform mesh in both x and y directions. The nonuniform mesh covers a region of $14.45C$ from the domain centre in each direction. The cell size of our nonuniform mesh varies according to a geometric progression with increasing cell size as we move away from the domain centre with 600 points in both x and y directions.

In Figure 12A, we show the axial velocity contours near the airfoil. We observe that the current UGKS-CGM solver satisfactorily captures smooth velocity profiles even when the body thickness reduces to almost 0 at the trailing edge.

In Figure 12B, we compare the coefficient of pressure (C_p) along the upper surface of the airfoil. We compare our simulation results with 2 different schemes, namely, the information preserving scheme⁴⁷ and NS solver with slip boundary conditions. The

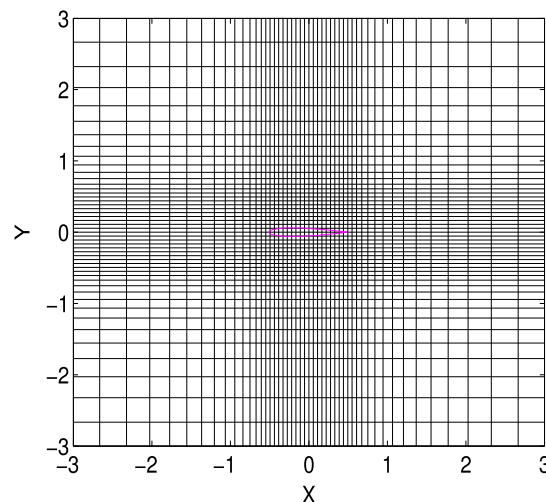


FIGURE 11 A portion of the meshed computational domain for flow past NACA 0012 airfoil (1 in every 10 mesh lines is shown for clarity) [Colour figure can be viewed at wileyonlinelibrary.com]

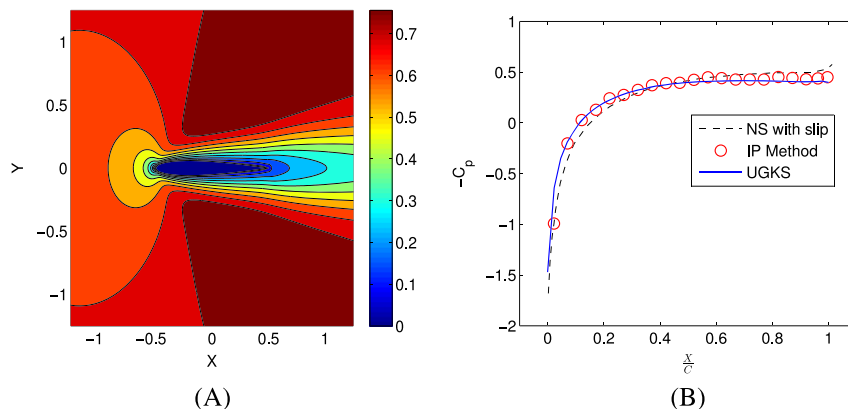


FIGURE 12 Simulation results for flow past NACA 0012 airfoil. A, Streamwise velocity contours in a small portion of the computational domain. B, C_p along the upper surface of airfoil. Here, IP Method refers to information preserving scheme. The data shown with the symbols have been extracted from Fan et al.⁴⁷ NS, Navier-Stokes; UGKS, unified gas kinetic scheme [Colour figure can be viewed at wileyonlinelibrary.com]

IP scheme is a variant of DSMC and is expected to be more accurate than the NS simulation with slip boundary condition. Data for comparison has been sourced from Fan et al.⁴⁷ We observe that the UGKS-CGM results are in close agreement with the IP method. It can also be observed that UGKS provides an improvement over the NS solver with slip boundary conditions. This demonstrates that the UGKS-CGM is indeed capturing non-NS physics and is also able to simulate non-Cartesian boundaries. All the above cases demonstrate that the UGKS-CGM method is quite versatile and works for a variety of boundary conditions, geometries, and Knudsen and Mach numbers.

4.5 | Criteria for determining the radius of the interpolation circle

Our interpolation procedure described in Section 3.3 requires a user-specified interpolation radius. In all the above test cases, we used an interpolation radius of $5\Delta x$. While this is above the minimum value that we specified in Section 3.3.1, the

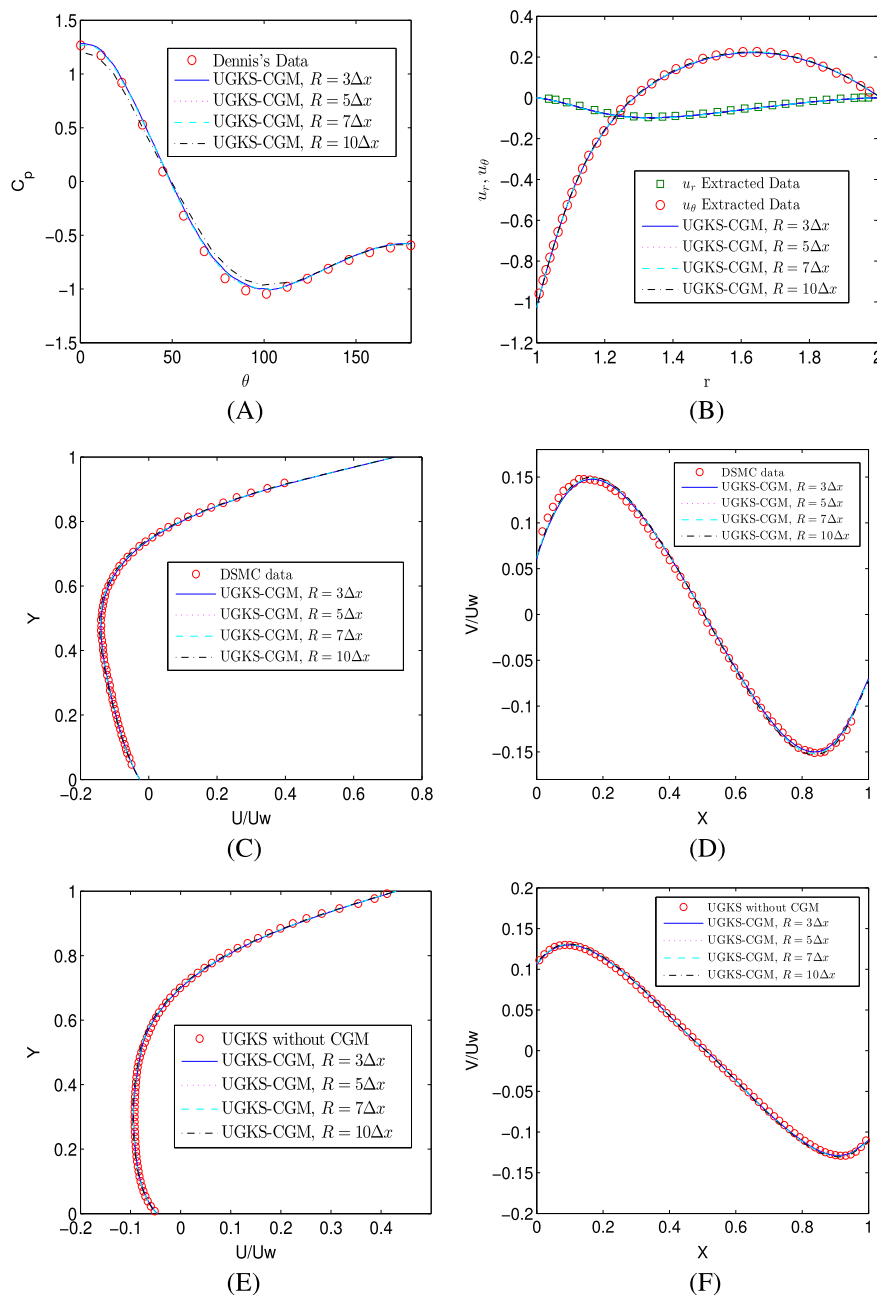


FIGURE 13 Effect of circle radius on steady state solution of A, flow past cylinder; B, polar lid driven cavity; C, u velocity at the vertical centreline and D, v velocity at the horizontal centreline of the rotated cavity in slip regime E, u velocity at the vertical centreline and F, v velocity at the horizontal centreline of the rotated cavity in transition regime. Here, Δx represents the minimum cell spacing used in the corresponding mesh. CGM, Cartesian grid method; UGKS, unified gas kinetic scheme [Colour figure can be viewed at wileyonlinelibrary.com]

maximum value of the radius is still subject to some uncertainty. In this section, we test the sensitivity of our flow field results to interpolation radius and determine, empirically, some criteria for the interpolation radius.

For the present study, we consider 4 test cases representing various geometries and Knudsen numbers. These are flow past a circular cylinder, polar lid-driven cavity, and rotated cavity in slip and transition regimes. For each case, we consider 4 different values of circle radius. We vary our circle radius in these simulations according to $R = n\Delta x$ where $n = 3, 5, 7$, and 10 for the present study and Δx represents the minimum cell size used in the corresponding simulation.

We first consider the effect of circle radius on a steady-state solution of flow past cylinder in the continuum regime (described in Section 4.3.1). In Figure 13A, we plot the variation of C_p versus θ for 4 different values of circle radius. We observe that the current solver performs well for small values of circle radius ($R = 3\Delta x, 5\Delta x$, and $7\Delta x$). However, when the circle radius is increased to $R = 10\Delta x$, we obtain a significant deviation between Dennis's results⁴⁵ and our UGKS-CGM simulation results.

In Figure 13B, we repeat the procedure for a polar lid-driven cavity (described in detail in Section 4.2). We observe that there is a very small deviation in the steady-state solution as we increase the circle radius. However, for a large value of the interpolation radius ($R = 10\Delta x$), we observe a tendency to drive the solution away from the correct results.

Finally, we consider the effect of circle radius on rotated cavity in the slip and transition regimes (described in detail in Section 4.1). We plot the normalised U and V velocities at the horizontal and vertical centreline of the cavity box for slip regime in Figure 13C and 13D and for transition regime in Figure 13E and 13F, respectively. Similar to the earlier cases, we observe that the circle radius has a minor effect on the steady state solution for small radii. However, larger values of circle radius ($R = 10\Delta x$) indeed show some deviations between the DSMC results and the current simulation results.

One can arrive at several other conclusions from the above study. Firstly, we can conclude that the UGKS-CGM solver gives accurate results for low values of circle radii (from $3\Delta x$ to $7\Delta x$). Secondly, these results are not only accurate but also robust, in that they are insensitive to changes in the interpolation radius. Finally, values of the interpolation radius beyond a certain limit (approximately $9\Delta x$) result in reduction of accuracy. This might be due to multiple reasons including the use of uniform weights for all points or due to the interpolation radius becoming comparable to the length scale of the immersed boundary. Our overall recommendation is that the interpolation radius be kept between $5\Delta x$ and $7\Delta x$ as this reduces computational effort and also retains robustness and accuracy.

5 | CONCLUSIONS

A novel technique, UGKS-CGM, to compute flows ranging from continuum to transition regime using a Cartesian mesh is developed in this paper. The new technique is capable of simulating viscous, incompressible, and moderately compressible flows from continuum to transition flow regime. The CGM developed in this paper differs from the usual, continuum CGM in 2 ways. Firstly, to capture non-equilibrium effects, the CGM used here interpolates both the microscopic distribution function and the macroscopic variables. Secondly, even for the macroscopic variables, a GKM-based density interpolation procedure is used at the boundaries, which allows the CGM to interface well with the UGKS method.

Extensive validation was performed by simulating both internal and external flows at various Knudsen numbers. The numerical results demonstrate the capability of the method to seamlessly compute flows with varying geometries, Mach and Knudsen numbers. Although the present solver can simulate flows up to transition regime, there is much scope of improvement in the present technique. Firstly, the method needs to be extended to flows with high Mach numbers and high Knudsen number flows. This would probably require the development of a robust weighted least squares procedure. Another problem that needs addressing is the efficiency of the base UGKS solver. While this is addressed to an extent through parallelisation, it is possible to address it through better integration schemes in phase space. Both these issues will be addressed in our future work.

ACKNOWLEDGEMENTS

This research work was supported by Department of Science and Technology (DST), New Delhi, India under the INSPIRE Fellowship scheme with fellowship grant number IF140144. Authors would also like to thank the IIT Delhi HPC facility for providing the computational resources. Authors would also like to thank Dr. Manish Agarwal, Computer Services Center, Indian Institute of Technology-Delhi for providing the necessary help in parallelising the code.

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How to cite this article: Ragta LK, Srinivasan B, Sinha SS. Unified gas kinetic scheme combined with Cartesian grid method for intermediate Mach numbers. *Int J Numer Meth Fluids*. 2017;85:507–524. <https://doi.org/10.1002/fld.4393>